University of Mosul
College of Computer Sciences and Mathematics
Computer Science Dept. /First Class

Principles of Computer Organization

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presentation notes

Lec. 4

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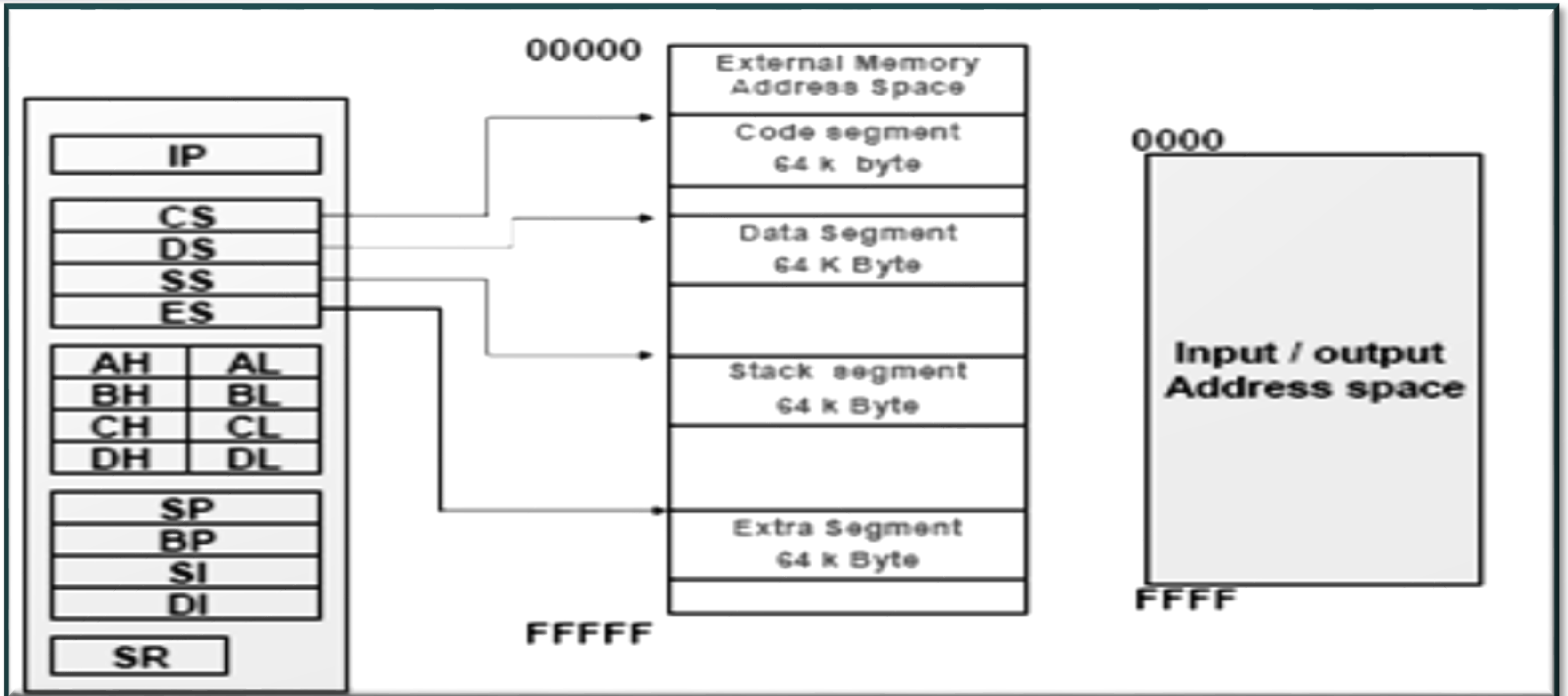
Evolution of 80X86 Family

- 8086, in 1978
  - First 16-bit microprocessor
  - 20-bit address bus, i.e. 2^{20} = 1MB memory
  - First pipelined microprocessor
- 8088
  - Data bus: 16-bit internal, 8-bit external
- 80286, 80386, 80486
  - Real/protected modes
  - Virtual memory

Software Model of the 8086 Microprocessor



- 8088/8086 microprocessor includes 13 16-bit internal registers:
  - Instruction pointer (IP)
  - Data registers (AX, BX, CX, and DX)
  - Pointer registers (BP and SP)
  - Index registers (SI and DI)
  - *Segment registers* (CS, DS, SS, and ES).
  - Status register (SR) or (Flag register)

Software Model of the 8086 Microprocessor

- 8086 architecture implements independent memory and input/output address spaces.
- The memory address space is 1,048,576 bytes (1 M byte) in length and the I/O address space is 65,536 bytes (64 Kbytes) in length.

Memory Address Space and Data Organization

- The 8086 microprocessor **supports 1Mbytes** of memory.
- This memory space is organized from a software point of view as **individual bytes** of data stored at consecutive addresses over the address range **00000_{16} to $FFFFF_{16}$**.
- The memory in an 8086-based microcomputer is actually organized as 8-bit bytes, not as 16 bit words.
- The 8086 can access **any two consecutive bytes as a *word* of data.**
- The lower address byte is the least significant byte of the word and the higher address byte is its most significant byte.

Memory Address Space and Data Organization

Example: From figure (a)

These two bytes represent the word $0101010100000010_2 = 5502_{16}$.

The high byte

low byte

The lower address byte (00724)
The higher address byte (00725)

Address	Memory (hexadecimal)
	:
00724	02
00725	55
	:

Figure (a)

LOW in LOW
HIGH in HIGH

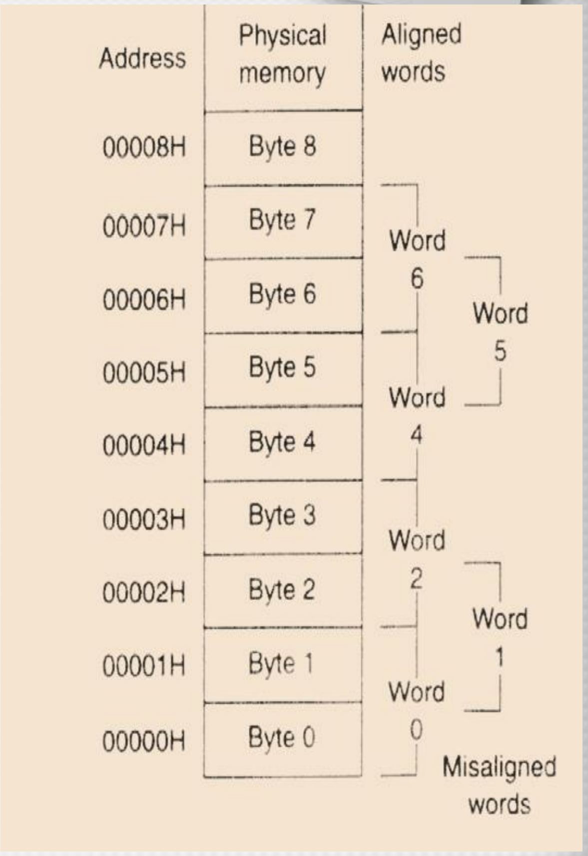
Memory Address Space and Data Organization

- To permit efficient use of memory, words of data can be stored at what are called even- or odd-addressed word boundaries.

- The least significant bit of the address determines the type of *word boundary*. If this bit is 0, the word is said to be held at an *even-address boundary*.

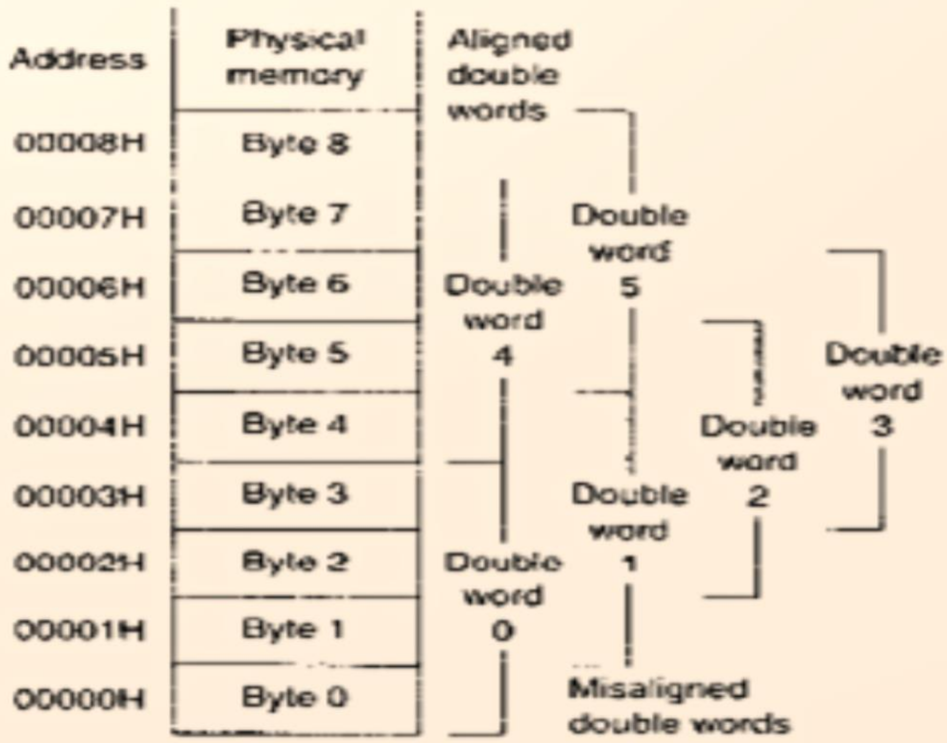
- A word of data stored at an even-address boundary, such as 00000_{16}, 00002_{16}, 00004_{16}, and so on, is said to be an aligned word. (all aligned words are located at an address that is a multiple of 2).

- A word of data stored at an odd-address boundary, such as 00001_{16}, 00003_{16}, or 00005_{16} and so on, is called misaligned word.



Memory Address Space and Data Organization

- A *double word* corresponds to four consecutive bytes of data stored in memory.



Memory Address Space and Data Organization

Example1:

Show how the word of data (02ED) store in memory location starting at address (EF05D)? Is the word aligned or misaligned?

02ED WORD , START ADDRESS = FE05D

MISALIGNED WORD.

	:
FE05D	ED
FE05E	02
	:

Example2:

Show how the double word of data (FF24EB55) store in memory location starting at address (A000E)? Is the double word aligned or misaligned?

WORD 2= FF24 WORD1=EB55

START ADDRESS A000E

	:
A000E	55
A000F	EB
A0010	24
A0011	FF
	:

Example3:

You have the following figure; What is the value of the word stored in memory starting at address (0210C)? Is the word aligned or misaligned?

0210B	:
0210C	22
0210D	0F
0210E	5A
0210F	00
02110	33
	:

WORD OF DATA = 0F22

0F	22
0210D	0210C

ALIGNED WORD

Example4:

You have the above figure; What is the value of the double word stored in memory starting at address (0210C)? Is the double word aligned or misaligned?

FIRST BUS CYCLE

WORD 1 (0F22)

SECOND BUS CYCLE

WORD2 (005A)

DOUBLEWORD= 005A0F22